Introduction to Artificial Intelligence

Shaping the Future with Intelligent Machines





What is AI?

Mimicking Intelligence

AI makes computers think and learn like humans, enabling them to process information and make decisions.

Problem Solvers

AI enables machines to solve complex problems, from playing chess to diagnosing diseases.

The New Electricity

As Andrew Ng said, "AI is the new electricity," powering innovation everywhere.



Rewind to the past

🧠 1. The Beginning (1940s–1950s)

- **1943: First concept of a neural network – McCulloch & Pitts model.**
- **1950: Alan Turing proposes the Turing Test to evaluate machine intelligence.**
- **1956: The term "Artificial Intelligence" is coined at the Dartmouth Conference by John McCarthy — the official birth of AI.**

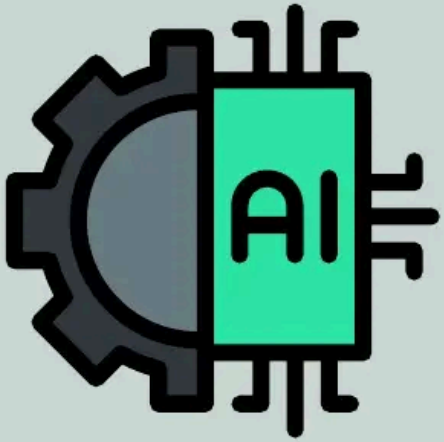
Rewind to the past

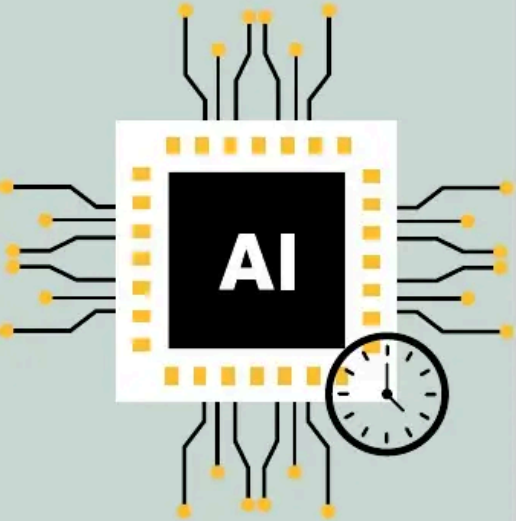
🔍 2. Early Progress (1956–1970s)

- Early programs could solve algebra problems, play checkers, and prove theorems.
- 1966: ELIZA – the first chatbot (simulating a psychotherapist).
- Limited computing power held back major breakthroughs.

🌧️ 3. AI Winter (1974–1980s)

- Hype didn't match performance → Funding cuts.
- AI research slowed due to high expectations and poor results.





Rewind to the past

🚀 4. Revival & Expert Systems (1980s)

- Rise of Expert Systems (e.g., MYCIN for medical diagnosis).
- Businesses began adopting AI for decision-making.

🔁 5. Second AI Winter (Late 1980s–1990s)

- Expert systems failed to scale.
- Another round of disappointment led to funding reductions.



Rewind to the past

💡 6. Machine Learning Era (2000s–2010s)

- Big data + better hardware → rise of Machine Learning (ML).
- AI outperforms humans in narrow tasks (e.g., Deep Blue beats chess champion in 1997).
- 2012: Deep Learning breakthrough with ImageNet.



Rewind to the past

🤖 7. Modern AI (2010s–Present)

- **Natural Language Processing (NLP): ChatGPT, Siri, Alexa**
- **Vision AI: Self-driving cars, face recognition**
- **AI in daily life: Healthcare, education, entertainment**
- **Major players: Google DeepMind, OpenAI, Meta AI, etc.**

🔮 8. The Future of AI

- **Focus on ethical AI, AGI (Artificial General Intelligence), and human-AI collaboration.**
- **AI is rapidly shaping every field from science to art.**





🚀 Scope of AI

🤖 Automation of Tasks

- AI can perform repetitive and complex tasks with high accuracy.
- Used in industries like manufacturing, customer service, and finance.

🧠 Smart Decision-Making

- AI helps in analyzing data and making decisions faster than humans.
- Widely used in stock trading, supply chain, and medical diagnosis.





🚀 Scope of AI

🗣️ Natural Language Processing (NLP)

- Enables machines to understand, interpret, and respond to human language.
- Applications: Chatbots, virtual assistants, language translation.

👁️ Computer Vision

- Allows machines to see and understand images and videos.
- Used in face recognition, medical imaging, and autonomous vehicles.

🚀 Scope of AI

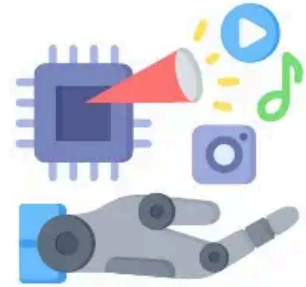
📊 Data Analysis & Predictive Modeling

- AI identifies patterns and trends in large datasets.
- Used in marketing, weather forecasting, sports analytics, etc.



🎮 Entertainment & Personalization

- AI powers recommendation systems (e.g., YouTube, Netflix).
- Creates personalized content and gaming experiences.



🚀 Scope of AI

🏥 Healthcare Transformation

- AI aids in early disease detection, robotic surgeries, and drug discovery.



📚 Education & E-learning

- Personalized learning paths using AI.
- Smart tutoring systems that adapt to a student's pace and style.



🚀 Scope of AI

🚗 Autonomous Systems

- Self-driving cars, drones, and robotics rely heavily on AI.
- The future of transportation and delivery services.

🌍 Global Challenges & Social Good

- AI is helping solve global problems: climate modeling, wildlife monitoring, disaster response.
- Assists NGOs and governments in making informed decisions.







Real-World Examples of AI

Recommendations

Netflix and Spotify use AI to suggest what you might like next.

Virtual Assistants

Siri, Alexa, and Google Assistant understand your voice commands.

Self-Driving Cars

Tesla and Waymo use AI to navigate roads safely and efficiently.

Image Recognition

Your phone uses AI for facial recognition and organizing photos.

Types of AI

Based on Capabilities

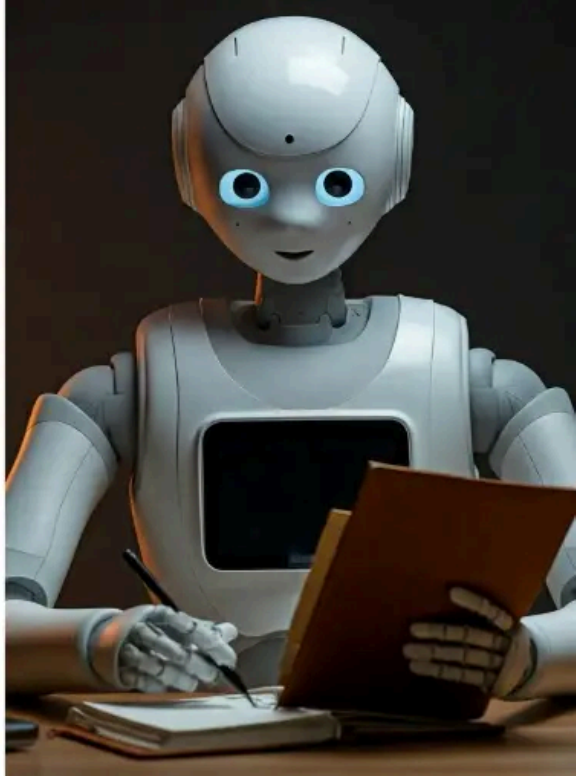
- Narrow AI (Weak AI)
- General AI (Strong AI)
- Super AI



Types of AI

Based on Functionality

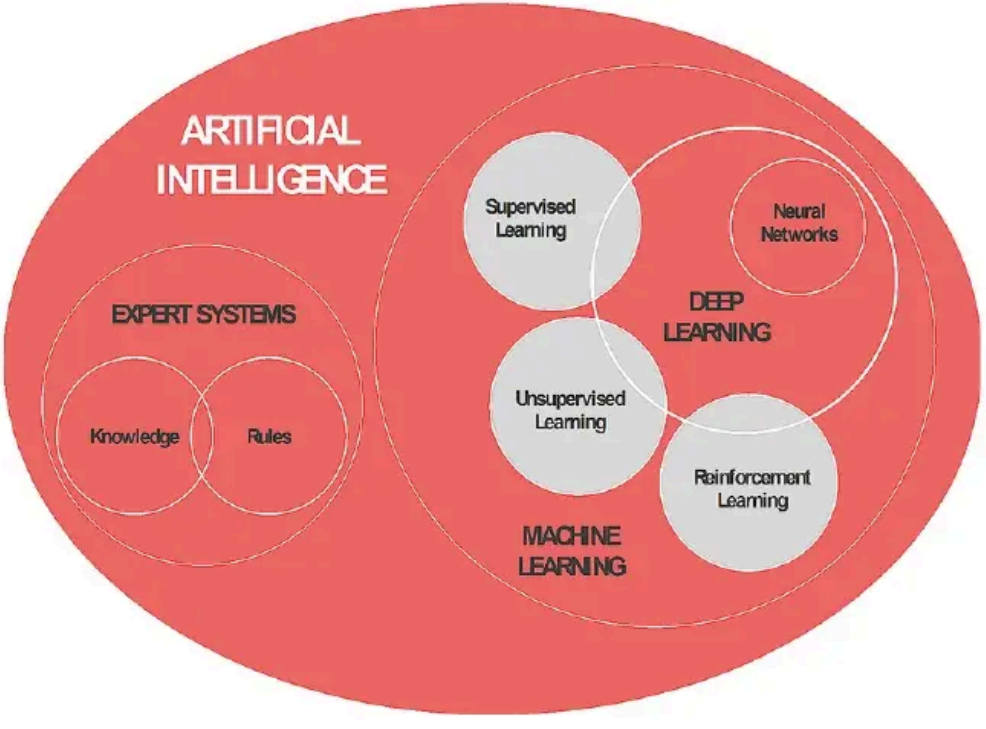
- Reactive Machines
- Limited Memory
- Theory of Mind
- Self-Aware AI

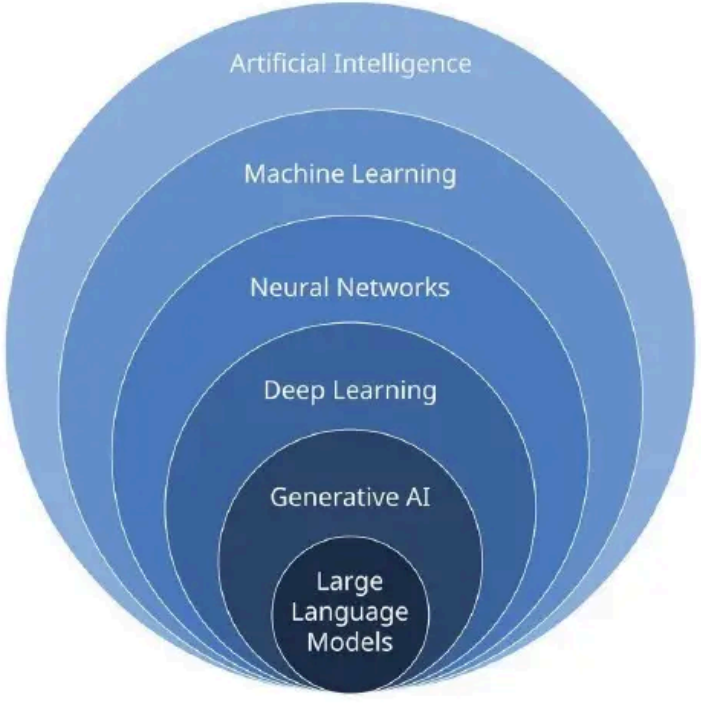


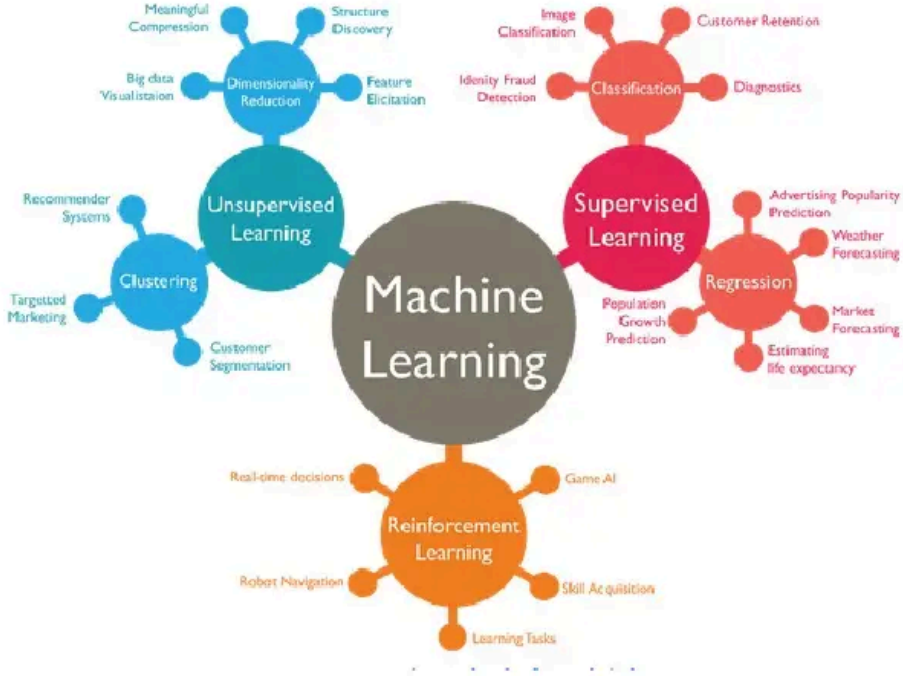
Branches of AI

- Machine Learning (ML)
- Deep Learning
- Natural Language Processing (NLP)
- Computer Vision
- Expert Systems
- Robotics
- Fuzzy Logic
- Reinforcement Learning

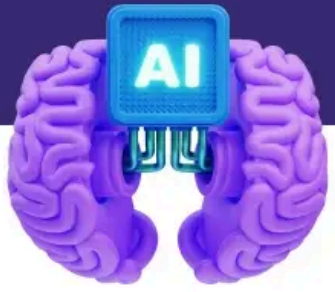



ARTIFICIAL INTELLIGENCE
EXPERT SYSTEMS
Knowledge
Rules
MACHINE LEARNING
Supervised Learning
Neural Networks
DEEP LEARNING
Unsupervised Learning
Reinforcement Learning


Artificial Intelligence
Machine Learning
Neural Networks
Deep Learning
Generative AI
Large Language Models


Machine Learning
Unsupervised Learning
Dimensionality Reduction
Meaningful Compression
Structure Discovery
Big data Visualisation
Feature Elicitation
Clustering
Recommender Systems
Targetted Marketing
Customer Segmentation
Supervised Learning
Classification
Image Classification
Customer Retention
Identity Fraud Detection
Diagnostics
Regression
Advertising Popularity Prediction
Weather Forecasting
Market Forecasting
Population Growth Prediction
Estimating life expectancy
Reinforcement Learning
Real-time decisions
Game AI
Robot Navigation
Skill Acquisition
Learning Tasks

AI Tools



Machine Learning & Deep Learning Tools

- **TensorFlow (by Google)**
  Open-source ML framework used for training deep neural networks.
- **PyTorch**
  Flexible and beginner-friendly deep learning library.
- **Scikit learn**
  Widely used for classical machine learning algorithms (e.g., decision trees, SVM).
- **Keras**
  High-level API for building and training deep learning models (runs on top of TensorFlow).

AI Tools

📊 Data Science & Analysis Tools

- **Pandas**

  Library for data manipulation and analysis in Python.

- **NumPy**

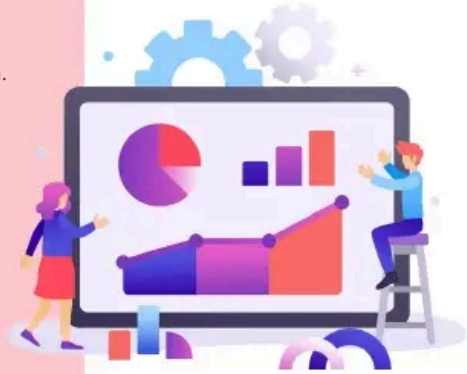
  Useful for handling large arrays and matrices of numeric data.

- **Matplotlib**

  Visualization library used to create charts and graphs.

- **Seaborn**

  Visualization library used to create charts and graphs.



AI Tools

🗣️ Natural Language Processing (NLP) Tools



- **spaCy**

  Fast NLP library for processing and understanding text.



- **NLTK (Natural Language Toolkit)**

  Useful for academic and research-based text processing.



- **Hugging Face Transformers**

  Pre-trained models for tasks like translation, summarization, and Q&A.



AI Tools

👁 Computer Vision Tools

- **OpenCV**

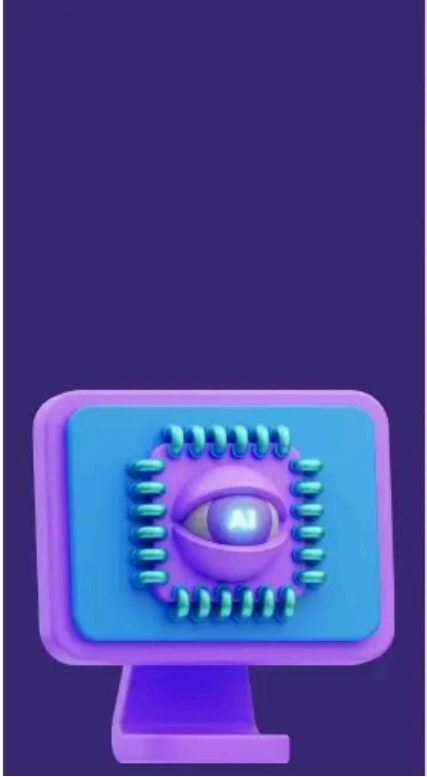
  Powerful library for image and video processing.

- **YOLO (You Only Look Once)**

  Real-time object detection system.

- **MediaPipe**

  Google's tool for face, hand, and pose detection (used in AR apps).



AI Tools

General AI Tools & Platforms



- **ChatGPT / GPT-4**
  Natural language generation and conversational AI (OpenAI).
- **Google Gemini**
  Google's conversational AI assistant.
- **Teachable Machine**
  Google's web-based tool to train simple models with no coding.
- **IBM Watson**
  AI platform for enterprise-level applications (e.g., healthcare, finance).



The Future of AI

Economic Growth

AI is projected to add trillions to the global economy by 2030.

Rapid Growth

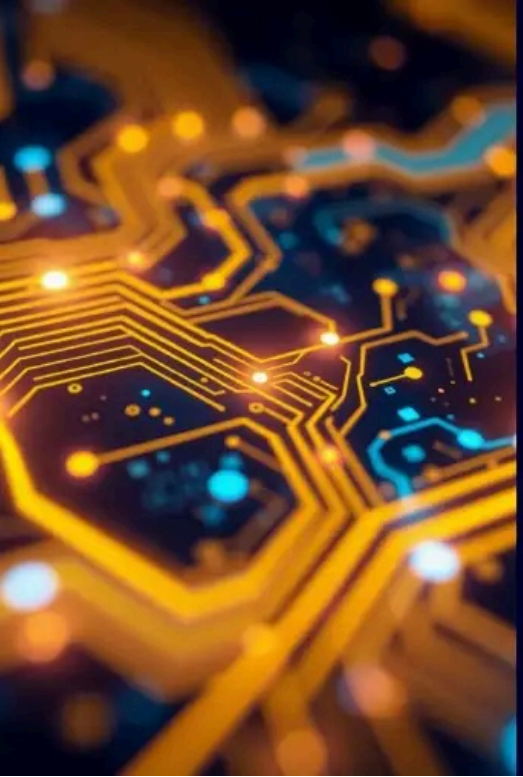
The AI market is expanding at a significant compound annual growth rate.

New Advancements

Expect continued breakthroughs in machine learning capabilities.

Industry Transformation

AI will reshape virtually every industry in the coming years.



Components of Artificial Intelligence

To build intelligent systems, AI combines multiple fields. These are called the components of AI

— they each focus on different abilities of “intelligence.”

What Are the Components of AI?

Artificial intelligence is a complex field made up of several interconnected components. These elements work in harmony to create intelligent systems capable of performing tasks that typically require human intellect. Each component is crucial for developing robust and effective AI solutions.

Interconnected Systems

AI components are not isolated; they work together to form cohesive and intelligent systems.

Unique Roles

Each component contributes a distinct and essential function to the overall development and operation of AI.

1. Learning

Machine learning is the backbone of modern AI, allowing systems to improve performance through data.

Supervised Learning: Training models with labeled datasets to make predictions (e.g., image classification).

Unsupervised Learning: Discovering patterns in unlabeled data (e.g., clustering customer segments).

Reinforcement Learning: Agents learn through trial and error by maximizing rewards in an environment (e.g., game AI).



2. Reasoning

Reasoning enables AI to draw conclusions and make inferences based on available information and established rules.



Deductive Reasoning

Deriving specific conclusions from general principles (e.g., if all birds lay eggs, and a robin is a bird, then a robin lays eggs).



Inductive Reasoning

Formulating general principles from specific observations (e.g., observing many white swans and concluding all swans are white).

3. Problem Solving

AI systems use various techniques to find solutions to complex problems, from simple puzzles to strategic challenges.



Maze Solving

Using algorithms to find the optimal path through intricate structures.

Game AI

Developing intelligent agents that can play and master complex games like chess or Go.

Search Algorithms

Algorithms like A* search efficiently explore possibilities to find the best solution.

4. Perception

Perception allows AI systems to interpret and understand sensory information from the real world.

Computer Vision

Enables AI to "see" and interpret visual data from images and videos

e.g., facial recognition, object detection in autonomous vehicles

Speech Recognition

Allows AI to "hear" and understand spoken language, converting audio into text

e.g., voice assistants, dictation software



5. Language Understanding (NLP)

Natural Language Processing (NLP) empowers AI to understand, interpret, and generate human language, bridging the gap between humans and machines.



Chatbots

AI-powered conversational agents that interact with users in natural language for customer service or information retrieval.



Translation

AI systems that can accurately translate text or speech from one language to another.



Sentiment Analysis

Identifying and extracting subjective information from text to determine emotional tone (e.g., positive, negative, neutral).

6. Knowledge Representation

This component focuses on how AI systems store, organize, and manage information to facilitate reasoning and understanding.

Rules

Representing knowledge as "if-then" statements to guide AI behavior and decision-making in expert systems.



Ontologies

Formal representation of knowledge as a set of concepts and their relationships within a domain, providing a structured vocabulary.



Knowledge Graphs

Networks of real-world entities and their relationships, allowing AI to understand complex connections and answer sophisticated queries.



7. Planning & Decision Making

AI systems utilize planning and decision-making capabilities to determine optimal sequences of actions to achieve specific goals.

Robot Navigation Developing algorithms that enable robots to autonomously navigate complex environments, avoiding obstacles and reaching destinations.

Task Scheduling Optimizing the allocation of resources and sequencing of tasks to maximize efficiency in various operations.

Search Trees Exploring different possible actions and their consequences to find the most effective path or solution.

8. Robotics & Action

This component brings AI into the physical world, enabling intelligent systems to perform actions and interact with their environment.



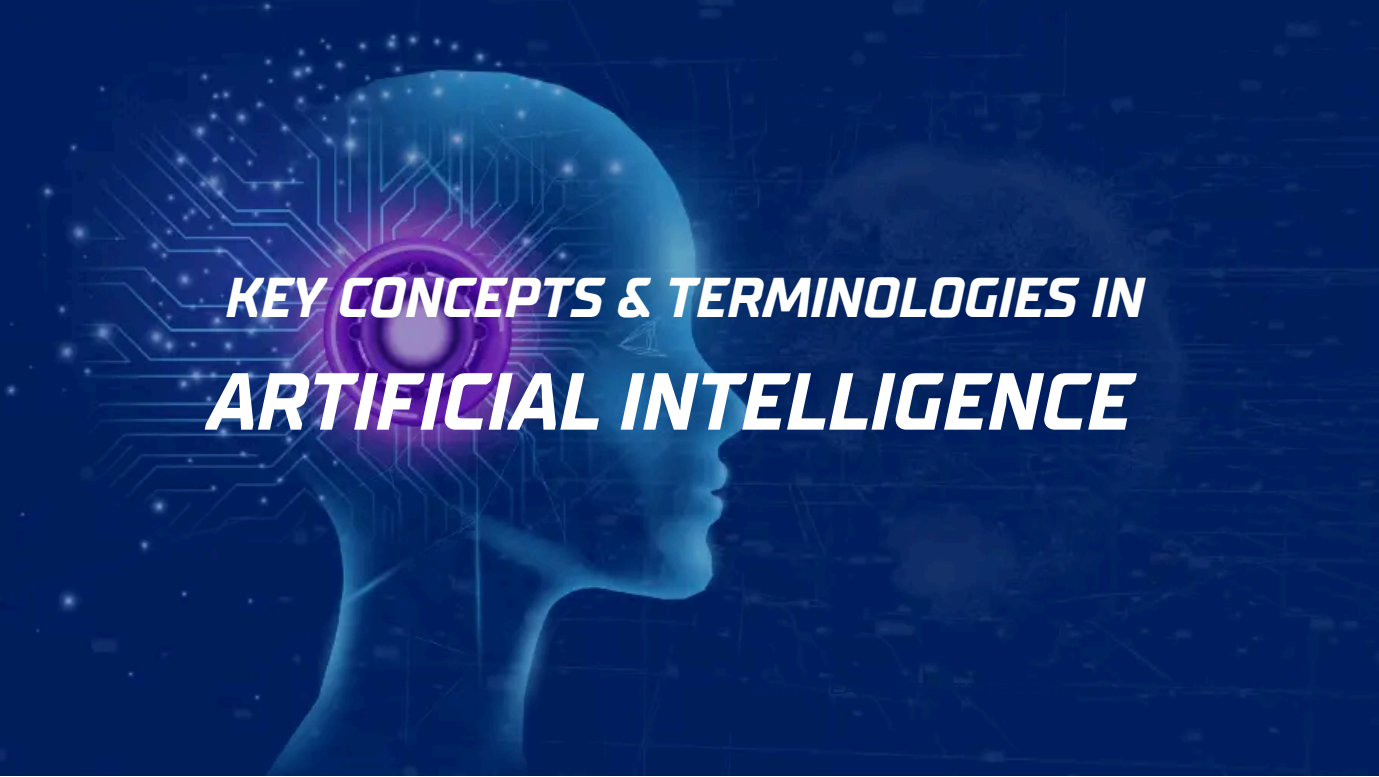
Humanoid Robots

Designing and programming robots that mimic human form and movements for various applications.



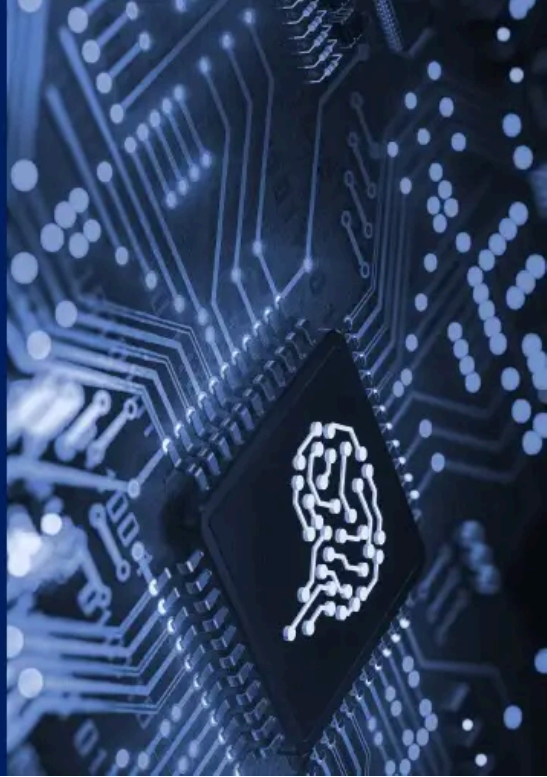
Self-Driving Cars

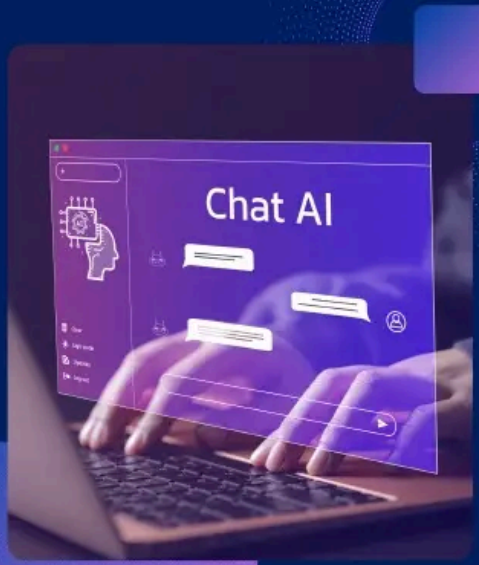
Integrating AI components like perception, planning, and control to enable autonomous vehicle operation.


KEY CONCEPTS & TERMINOLOGIES IN
ARTIFICIAL INTELLIGENCE

Key concepts

- Agent
- Backward Chaining
- Assisted intelligence
- Augmented intelligence
- Autonomous intelligence
- Machine learning
- Expert systems
- Neural networks
- Deep learning
- Reinforcement Learning

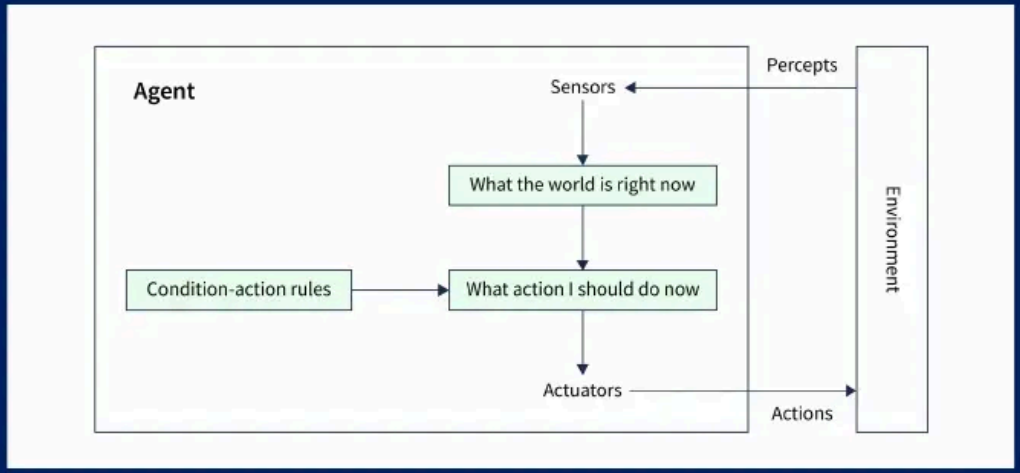




AGENT

- An agent is a entity that can function independently and act on behalf of a user or another program.
- An agent can perceive its surroundings, make choices depending on its condition and goals, and take action to accomplish those choices.
- Agents are frequently utilized in AI applications like robotics, games, and natural language processing.
- Depending on their level of intelligence, actions, and interactions with the environment, there are various types of agents.

AGENT


Agent
Sensors
Percepts
What the world is right now
Condition-action rules
What action I should do now
Actuators
Actions
Environment

AGENTS



SIMPLE REFLEX AGENTS

A simple reflex agent is a type of intelligent agent that makes decisions based only on the current percept, ignoring the rest of the percept history.



MODEL-BASED REFLEX AGENTS

These agents use an internal environment model to make informative decisions. They can handle more complex situations that require planning and reasoning.



GOAL-BASED AGENTS

Goal-based agents process the environment's state against the objectives they have set. These agents have been fed with explicit goals



UTILITY-BASED AGENTS

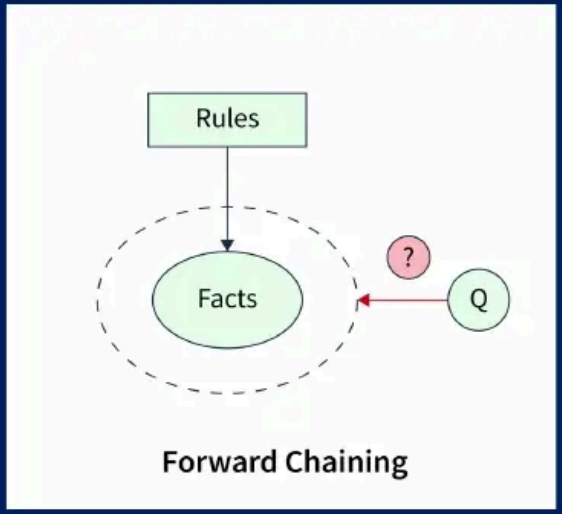
consider the costs and benefits of their actions. They choose activities that have the highest expected utility. They are suitable for decision-making under uncertainty



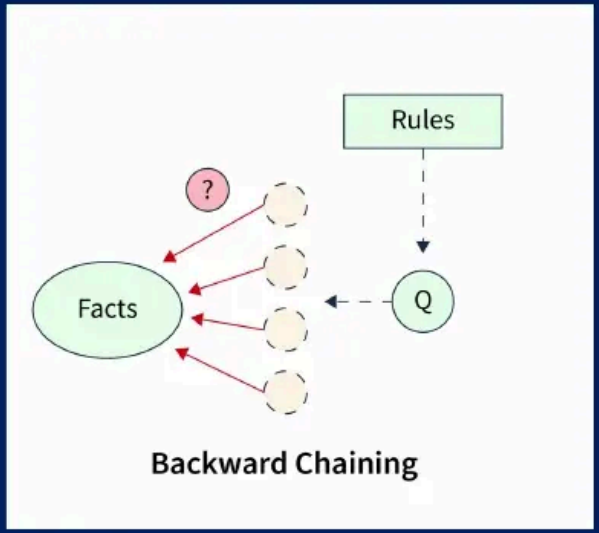
LEARNING AGENTS

As they gain knowledge from previous experiences, learning agents gradually become more effective.

FORWARD CHAINING



BACKWARD CHAINING



INTELLIGENCE



ASSISTED INTELLIGENCE

This is when AI acts as a tool to support human decision-making.



AUGMENTED INTELLIGENCE

This describes AI designed to enhance and work in partnership with human intelligence.

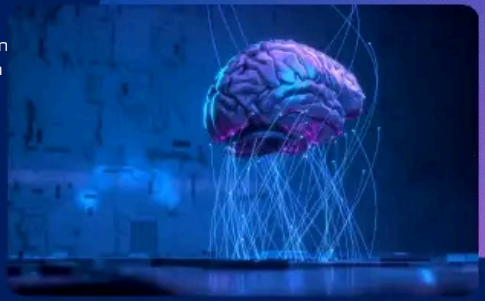


AUTONOMOUS INTELLIGENCE

This refers to AI systems that can operate and make decisions independently, without any human involvement.

MACHINE LEARNING

Machine learning is a type of AI where computers learn from data to perform tasks without being explicitly programmed. Instead of following rules written by a programmer, a machine learning model is trained on a dataset to find patterns, which allows it to make predictions or decisions on its own.



TYPES OF MACHINE LEARNING

- SUPERVISED LEARNING
- UNSUPERVISED LEARNING
- REINFORCEMENT LEARNING



EXPERT SYSTEMS

An expert system is an AI program designed to replicate the decision-making ability of a human expert in a specific field. It consists of two main parts: a knowledge base, which contains specialized facts and rules for a domain like medicine or finance, and an inference engine, which uses that knowledge to reason and draw conclusions.



KNOWLEDGE BASE

The knowledge base is the repository of information and rules for an expert system

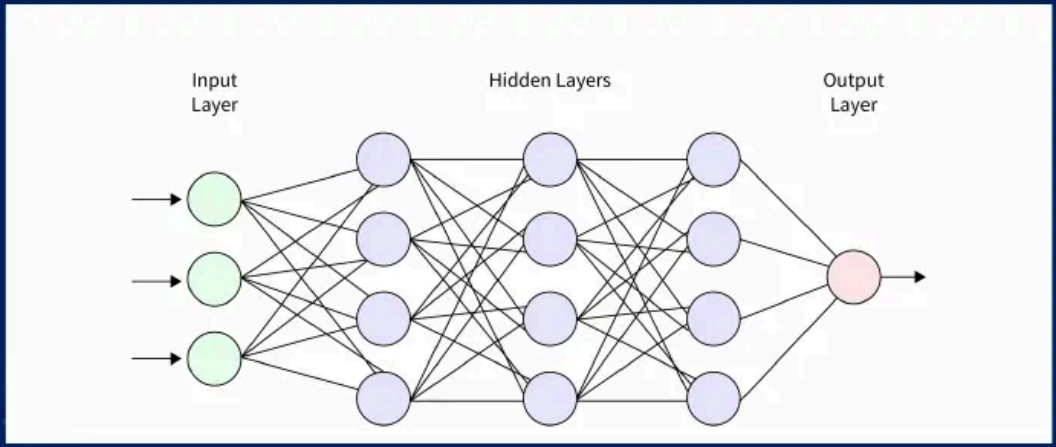


INFERENCE ENGINE

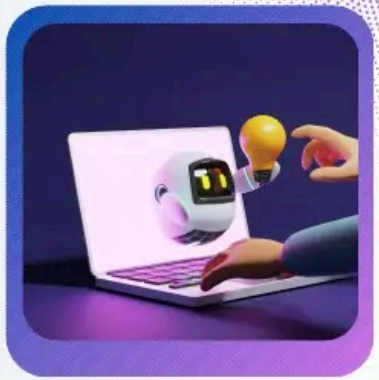
The inference engine is the "brain" or the reasoning component of the expert system.

NEURAL NETWORKS

A neural network, also known as an Artificial Neural Network (ANN), is a type of machine learning inspired by the human brain.



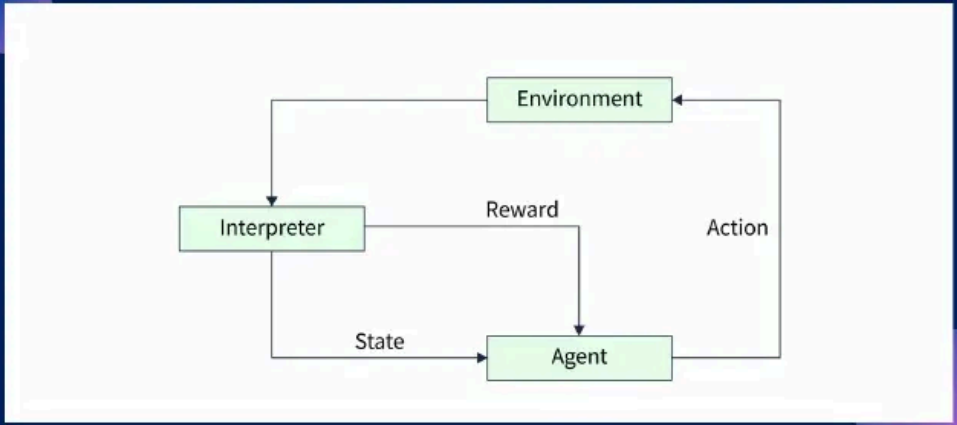
DEEP LEARNING



- Uses multiple-layered artificial neural networks to process and analyse very vast and complicated datasets.
- In deep learning, neural networks can learn and extract hierarchical representations of the input data because they are built from multiple layers of interconnected nodes.
- Beneficial in applications like image and audio recognition, natural language processing, and autonomous cars because they can learn and recognise patterns in the data.
- Deep learning algorithms are utilised in fraud detection, predictive maintenance, and recommendation systems.

REINFORCEMENT LEARNING

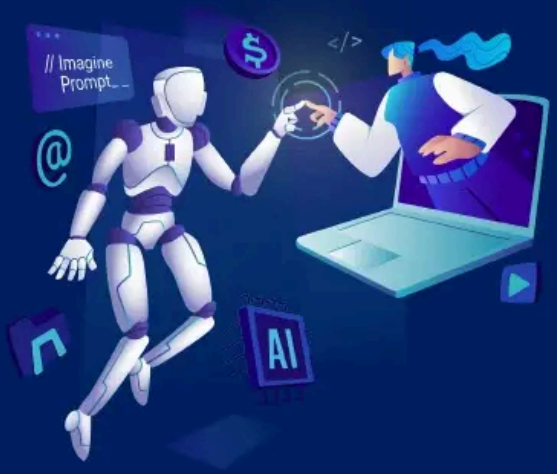
Reinforcement learning (RL) is an AI concept that enables an agent to learn from experience by interacting with an environment.



UNDERSTANDING AI TERMINOLOGY

Mastering the Vocabulary of Artificial Intelligence

- AI terminology defines the specialized vocabulary essential for understanding and working with Artificial Intelligence.
- These terms help us make sense of how machines simulate human intelligence.
- Knowing these terms enhances your ability to engage with AI systems and tools in real-world applications.



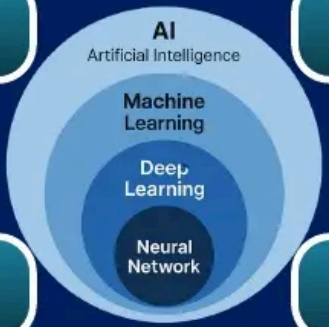
AI CORE CONCEPTS

Artificial Intelligence (AI)

Simulation of human intelligence by machines.

Machine Learning (ML)

Algorithms that improve through experience.



Neural Network

Modeled after the human brain to recognize patterns.

Deep Learning (DL)

Neural network-based learning models

TYPES OF AI SYSTEMS

- **Reactive Machines**
  No memory, reacts only to current input.
- **Limited Memory**
  Uses past experiences for decision-making.
- **Theory of Mind**
  Understands emotions and beliefs (still in research).
- **Self-Aware AI**
  Sentient machines (theoretical/future concept).

KEY COMPONENTS OF AI SYSTEMS

- **Algorithm**
  Step-by-step instructions for solving tasks.
- **Training Data**
  Data used to teach models.
- **Hyperparameter**
  Controls model training process.
- **Token**
  A chunk of input text processed by LLMs.

AI CAPABILITIES

Natural Language Processing

Understands human language



Computer Vision

Interprets visual data



Voice/Speech Recognition

Converts speech to text



Image Recognition

Identifies patterns in images



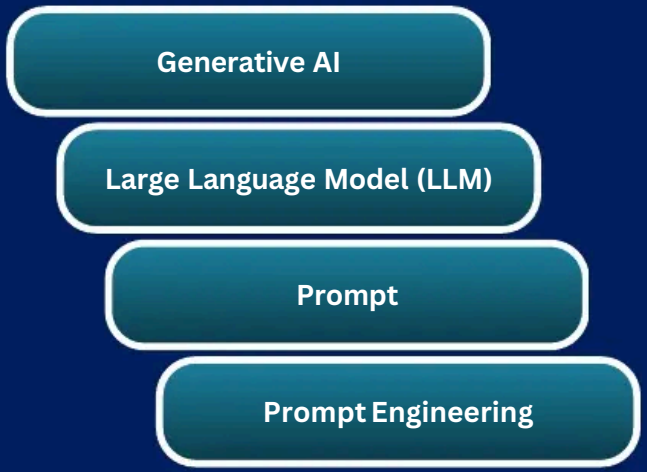
DATA TERMINOLOGY

Big Data	Massive, complex datasets
Structured Data	Organized in rows/columns
Unstructured Data	Text, images, audio—unorganized
Data Mining	Extracting patterns from data.

ADVANCED AI CONCEPTS

- **TRANSFER LEARNING**
  Applies knowledge to new tasks.
- **EMERGENT BEHAVIOR**
  Unexpected AI capabilities.
- **HALLUCINATION**
  AI outputs false but confident info.
- **GUARDRAILS**
  Rules to ensure safe AI use.

GENERATIVE AI & LLMS



PROBLEM SOLVING IN AI

An Introduction to Intelligent Agent-based Problem Solving



Key Topics

1. What is Problem Solving in AI?
2. Types of Problems AI Solves
3. Problem Formulation
4. State Space Search
5. Search Strategies
6. Example: **8-Puzzle, Water Jug, Maze Solving**
7. Real-world Applications



What is Problem Solving in AI?

Problem solving = finding a sequence of actions that lead to a goal

- Search :
  AI explores a problem space using search algorithms to find optimal or feasible solutions.
- Agent-based approach :
  An AI agent interacts with its environment by:
  - Perceiving the current state
  - Deciding what action to take
  - Acting to reach a goal



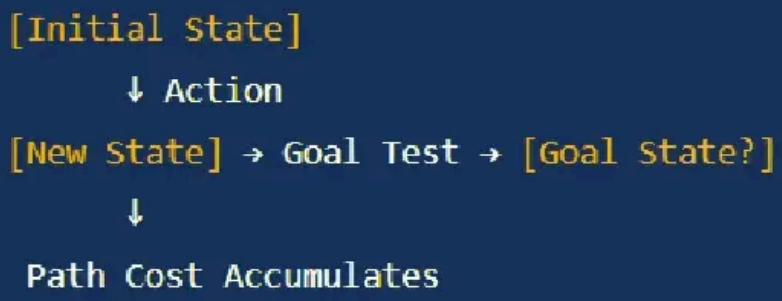
Characteristics of AI Problems

Characteristic	Description	Example
Well-Defined Problem	Clear goals, states, and rules for actions	Chess, 8-Puzzle, Sudoku
Ill-Defined Problem	Ambiguous goals or lack of clear structure	Story generation, conversation
Deterministic	Outcome of actions is predictable and fixed	Maze solving, robot in grid
Non-Deterministic	Outcomes of actions may vary, even for the same input	Self-driving in traffic
Static	Environment does not change while solving the problem	Crossword puzzle
Dynamic	Environment changes while the agent is planning/acting	Stock market prediction
Fully Observable	Agent has complete information about the environment	GPS navigation with satellite view

What is Problem Formulation?

Problem Formulation is the process of defining a problem in a way that an AI agent can understand and solve it using search.

Component	Description	Example (8-Puzzle)
Initial State (s_0)	Starting condition of the agent	Initial tile configuration
Actions (A)	All possible moves the agent can take	Move blank tile up, down, etc.
Transition Model (T)	Result of applying an action on a state	New puzzle state after move
Goal Test (G)	Checks if the current state is the goal	Is puzzle in goal config?
Path Cost (C)	Total cost from start state to goal (e.g., steps taken)	Number of moves made


[Initial State]
↓ Action
[New State] → Goal Test → [Goal State?]
↓
Path Cost Accumulates

What is State Space?

The State Space is the complete set of all possible states that can be reached by the agent through a sequence of actions.

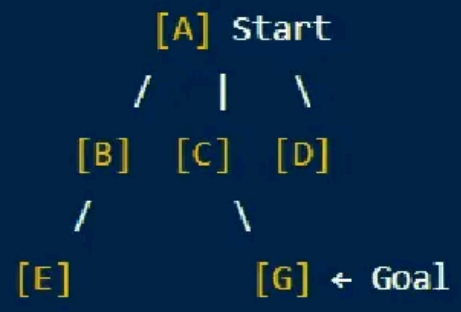
- Each state represents a specific situation or configuration of the problem.
- Initial State: The starting point of the agent.
- Goal State: The desired state the agent wants to reach.
- Actions: Moves that transition the agent from one state to another.
- The agent navigates through the state space to find a solution.

Example – 8 Puzzle

Search Tree

A Search Tree is a graphical representation of how an AI agent explores different possible paths from the initial state to reach a goal state.

- Each node in the tree represents:
  - A state in the state space
  - The path taken from the initial state
  - The action applied to reach it
- The root node is the initial state.
- Branches represent actions taken.
- Leaf nodes may be:
  - Dead ends (non-goal states)
  - Goal states (solutions)

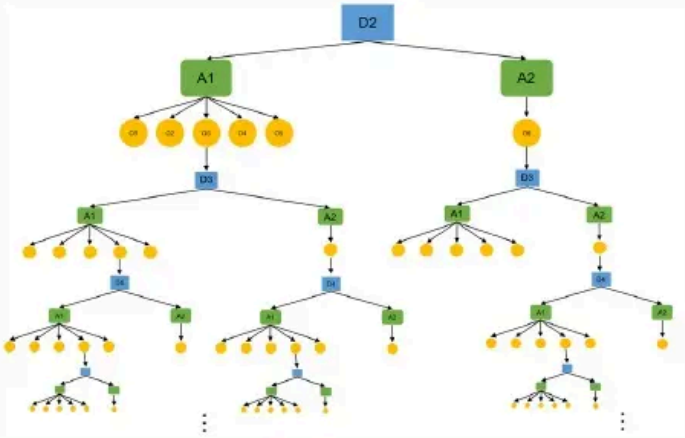


Nodes vs. States

Term	Meaning
State	A configuration of the problem (e.g., position in a maze)
Node	A data structure containing: state + path + cost + depth

What is a Solution?

A solution is a path from the initial state to a goal state through a sequence of valid actions.



Uninformed Search

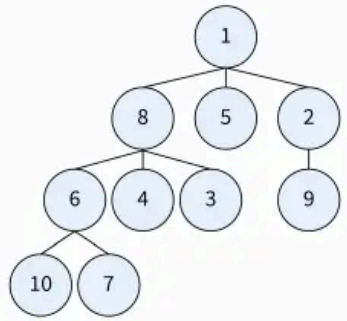
- These are search algorithms that do not use any domain-specific knowledge (heuristics).
- They explore the state space blindly, using only the problem definition.

Algorithm	Description
Breadth-First Search (BFS)	Explores nodes level by level (shallowest first)
Depth-First Search (DFS)	Explores deepest node first before backtracking
Uniform Cost Search	Expands node with the lowest path cost (not necessarily shortest path)
Depth-Limited Search	DFS with a depth cut-off to prevent infinite paths
Iterative Deepening DFS	Combines BFS's completeness with DFS's space efficiency by repeating DFS with increasing depth limits

Breadth-first Search

1. Start at node 1.
2. Visit all Level 1 nodes: 8, 5, 2.
3. Visit all Level 2 nodes: 6 (from 8), 4 (from 8), 3 (from 8), 9 (from 2).
4. Visit all Level 3 nodes: 10 (from 6), 7 (from 6).

Order of Visitation: 1 → 8 → 5 → 2 → 6 → 4 → 3 → 9 → 10 → 7



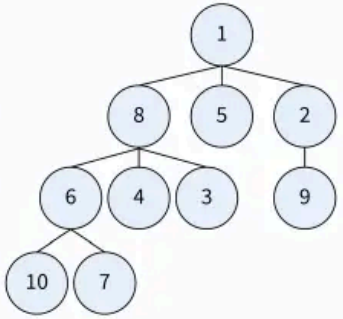
- **Advantages: Finds the shortest path; complete (will find the goal if it exists).**
- **Disadvantages: Requires more memory (stores all nodes at each level)**

Depth-First Search

1. Start at node 1.
2. Go to 8, then 6, then 10.
3. Backtrack to 6, visit 7.
4. Backtrack to 8, visit 4 and then 3
5. Backtrack to 1, go to 5.
6. Backtrack to 1, go to 2, then 9.

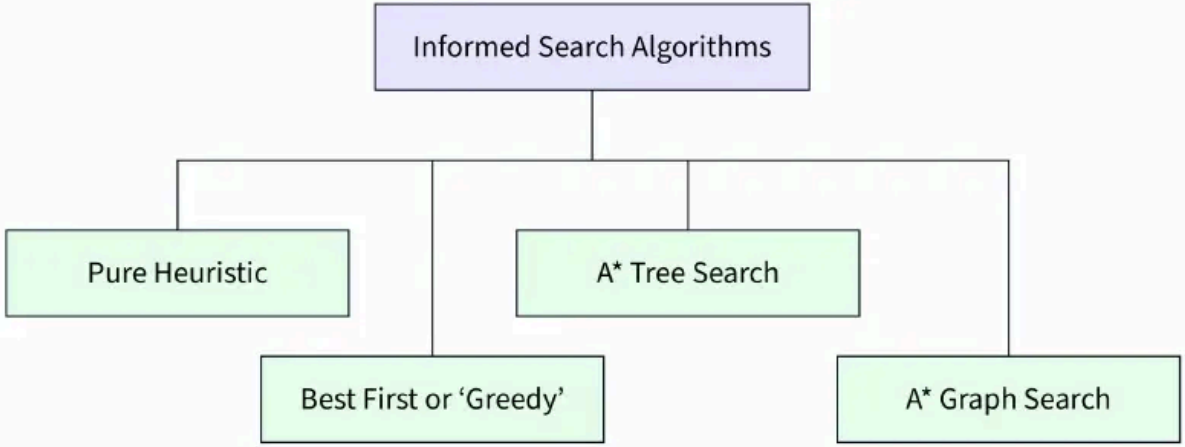
Order of Visitation: 1 → 8 → 6 → 10 → 7 → 4 → 3 → 5 → 2 → 9

- **Advantages: Uses less memory (only needs to store the current path).**
- **Disadvantages: May not find the shortest path; can get stuck in infinite loops in infinite trees.**



Informed Search

- Informed search algorithms are a type of search algorithm that uses heuristic functions to guide the search process.



Informed Search

Heuristic Function (h(n))

- A function that estimates the cost from node n to the goal.
- Good heuristics help reduce time and space complexity.
- Example: Straight-line distance in a map problem.

Evaluation Function (f(n))

Combines heuristic and path cost:

$$f(n) = g(n) + h(n)$$

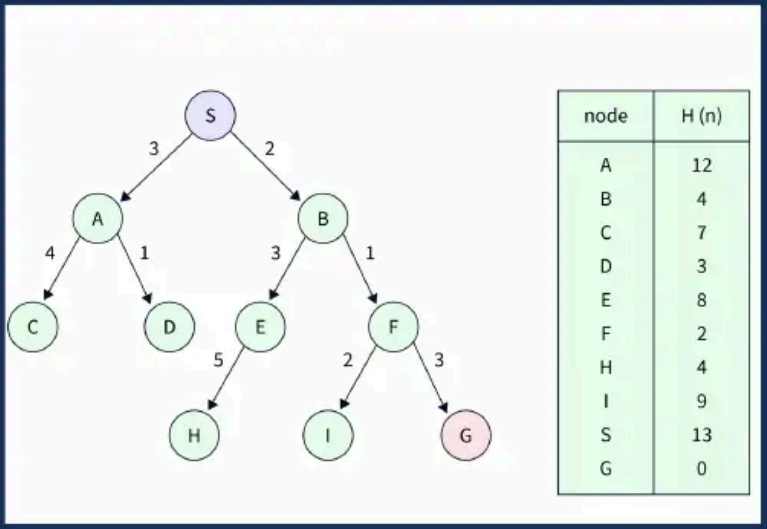
where:

g(n) : Cost to reach node n from the start.
h(n) : Estimated cost from n to the goal.

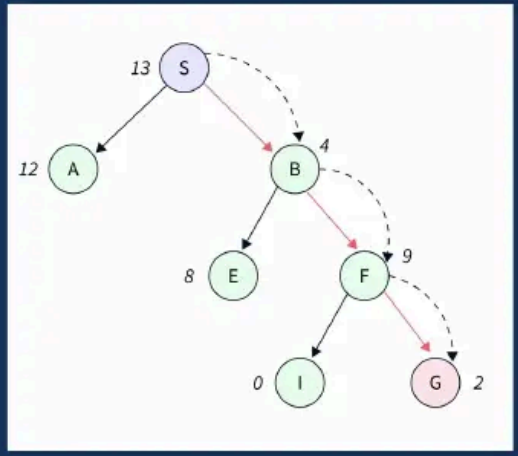
Best First Search Algorithm (Greedy search)

1.Start node added to the open list.
2.While the open list is not empty:
  a. Eliminate the node from the open list with the shortest calculated distance.
  b. Provide the solution if the node is the goal node.
  c. Generate the children of the current node.
  d. Estimate the children's travel time to the objective after adding them to the open list.
3. There is only a solution if the open list is empty.

Expands the node with the lowest heuristic value H(n), ignoring the path cost g(n). It's a heuristic-driven search that prioritizes nodes closest to the goal (estimated by H(n)).



node	H (n)
A	12
B	4
C	7
D	3
E	8
F	2
H	4
I	9
S	13
G	0



Path: S → B → F → G.

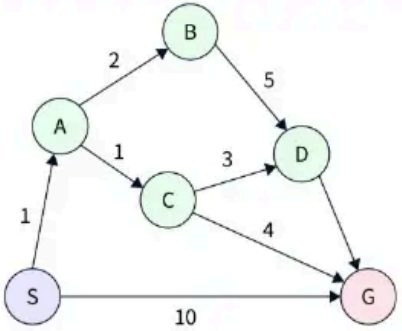
Total Cost: g(n)=2+1+3=6

Nodes Expanded: S, B, F.

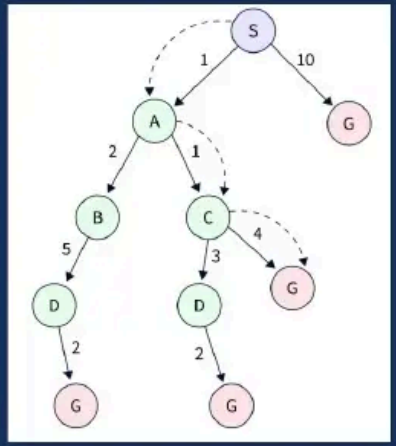
A* Search Algorithm

1. Set the starting node's g-value to 0 and its h-value to the expected cost of getting there from the starting node.
2. Expand the node with the lowest total f(n) cost.
3. Stop the search and return the route from the starting node to the goal node if the expanded node is the desired node.
4. If not, determine the g-value and h-value, and update the f-value for each neighbor of the expanded node. Finally, add the neighbor to the open list if it's not already there.
5. Until the goal node is located or the open list is empty, repeat steps 2-4.

Expands the node with the lowest f(n)=g(n)+H(n), balancing the cost to reach the node (g(n) and the estimated cost to the goal (H(n)



State	h (n)
S	5
A	3
B	4
C	2
D	6
G	0

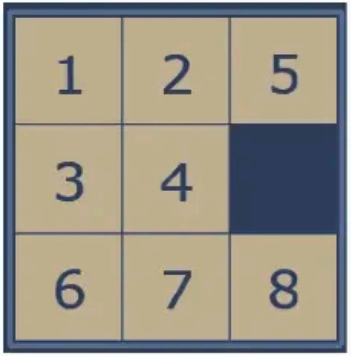


Path: S → A → C → G.

Total Cost: g(n)=1+1+4=6

Nodes Expanded: S, A, C

8-Puzzle Problem



🔹 **Initial Configuration (Start State)**

1	2	3
4	_	6
7	5	8

🎯 **Goal Configuration**

1	2	3
4	5	6
7	8	_

🔄 **Legal Actions**

The blank tile (_) can move:
Up - Down
Left - Right

🔁 **State Transitions**

Applying an action changes the state by swapping the blank tile with a neighboring tile

Problem Solving Examples

🪣 1. Water Jug Problem

Problem Statement :
Given two jugs with capacities of 4 liters and 3 liters, measure exactly 2 liters using only the jugs.

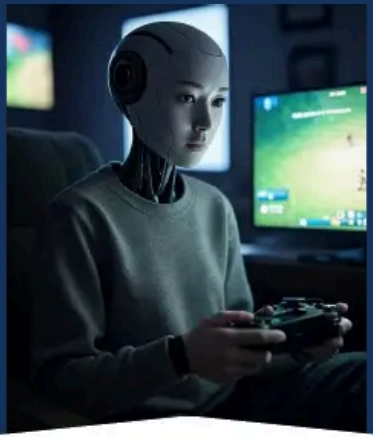
🌀 2. Maze Solving Problem

Problem Statement:
Navigate an agent from a start cell to a goal cell in a maze avoiding obstacles.

Real-World Applications of Problem Solving in AI



🤖 1. Pathfinding in Robotics



🎮 2. Game AI

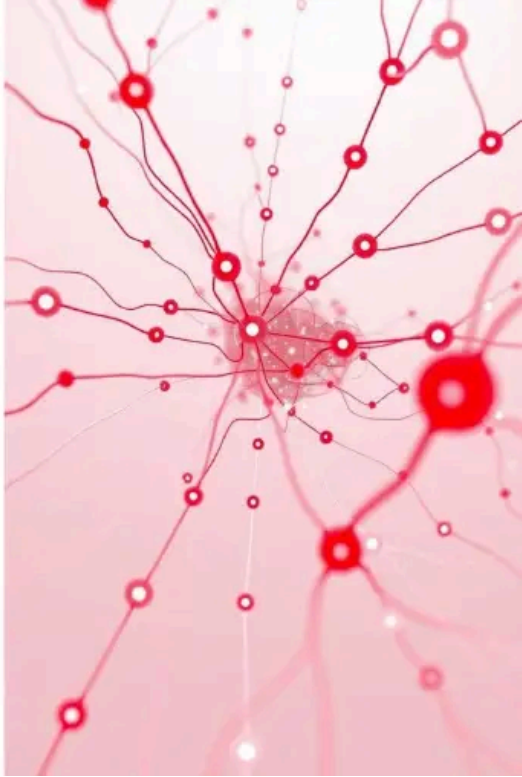


🗺️ 3. Navigation (GPS Systems)

KNOWLEDGE

REPRESENTATION IN AI

How AI stores, understands, and reasons with knowledge



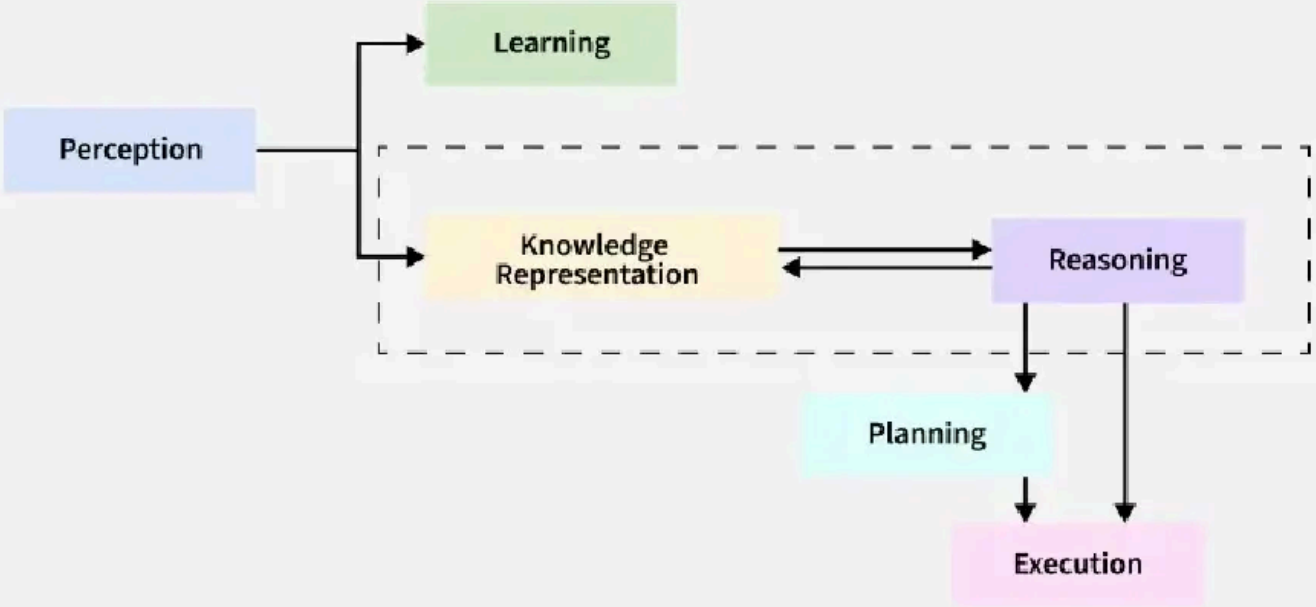
What is Knowledge Representation?

Enabling Intelligent Behavior

Helps AI reason, make decisions, and understand complex situations. It forms the basis for intelligent behavior in machines.

Bridging the Gap

Transforms human-understandable knowledge into a machine-readable format, crucial for advanced AI applications.


Learning
Perception
Knowledge Representation
Reasoning
Planning
Execution

Knowledge and Intelligence

Knowledge as a Foundation

Knowledge provides facts, rules, and data (e.g., traffic laws for self-driving cars).
Without it, intelligence lacks the raw material to act.

Intelligence as Application

Intelligence applies knowledge to solve problems (e.g., a robot using physics principles to navigate terrain)

Interdependence

Static knowledge becomes obsolete without adaptive intelligence. Conversely, intelligence without knowledge cannot reason or learn (e.g., an AI with no medical database cannot diagnose diseases)

Synergy

Effective AI systems merge robust knowledge bases (the what) with reasoning algorithms (the how). For example, ChatGPT combines vast language data (knowledge) with transformer models (intelligence) to generate coherent text

Why is Knowledge Representation Important?



Reasoning & Inference

Allows AI systems to derive new conclusions from existing knowledge.



Decision-Making

Enables informed choices by processing complex information and relationships.



Understanding Context

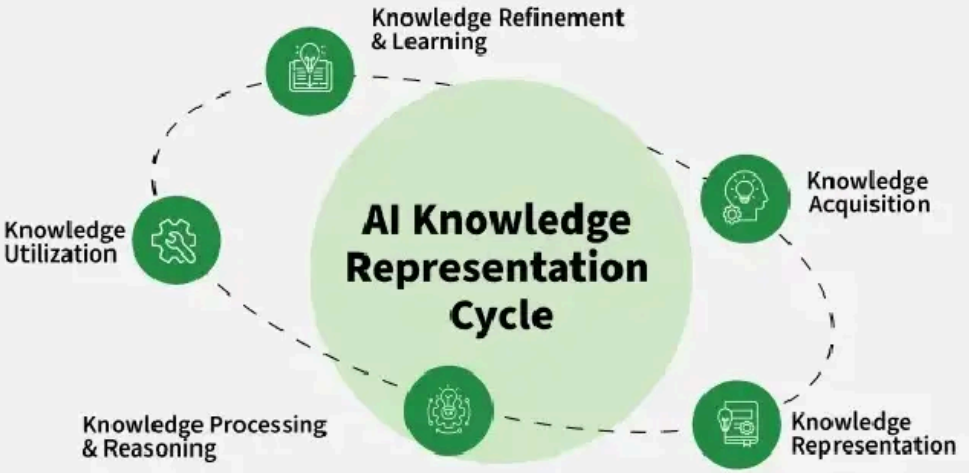
Provides the necessary framework for interpreting ambiguous or nuanced data.



Problem-Solving

Facilitates the systematic approach to complex challenges by organizing information.

The AI Knowledge Cycle



Knowledge Acquisition

AI systems collect data from various structured and unstructured sources.

Sources

Structured
Databases, spreadsheets
Unstructured
Text, speech, images, videos

Techniques Used

Machine Learning
Natural Language Processing (NLP)
Computer Vision



Knowledge Representation

The collected knowledge is transformed into structured formats suitable for reasoning and decision-making.

Forms of Representation

- Semantic Networks
- Frames
- Rules (IF-THEN)
- Ontologies
- Logic (Propositional & Predicate Logic)
- Probabilistic models (Bayesian Networks)



Knowledge Processing & Reasoning

AI applies logical and statistical techniques to manipulate and analyze knowledge.

- **Deductive & Inductive reasoning**
- **Problem-solving using heuristics**
- **Decision-making under uncertainty**

Techniques Used

- **Rule-based systems**
- **Deep learning models**
- **Reinforcement learning**



Knowledge Utilization

AI leverages its knowledge to perform meaningful tasks.

Forms of Representation

- **Chatbots: Understanding and responding to queries**
- **Recommendation Engines: Suggesting relevant content**
- **Autonomous Vehicles: Navigating in real-time**
- **Robots: Performing physical tasks**



Knowledge Refinement & Learning

Continuous learning and adaptation improve system performance over time.

Mechanisms

- **Supervised Learning: Updating based on labeled data**
- **Reinforcement Learning: Feedback-based adjustment**

Outcome

- **Improved accuracy**
- **Adaptability to new situations**
- **Reduced errors over time**



✅ Key Takeaways

The AI Knowledge Cycle is iterative and self-improving.

It’s central to the development of intelligent, autonomous systems.

Continuous feedback and learning make AI robust, scalable, and future-ready.



🔄 Process of Knowledge Representation

Acquisition – Getting knowledge from humans, text, or sensors

Representation – Encoding knowledge in structured form

Storage – Maintaining it in databases or knowledge bases

Retrieval – Accessing it when needed

Reasoning – Deriving new knowledge or making decisions



Types of Knowledge in AI

🔢 Type	📝 Description	💡 Example
1. Declarative Knowledge	Facts and static information	“Kerala is a state in India”
2. Procedural Knowledge	Instructions or steps for performing tasks	“How to drive a car”
3. Meta-Knowledge	Knowledge about knowledge (self-awareness)	Knowing that you know the capital of France
4. Heuristic Knowledge	Experience-based rules or best guesses	“If it’s cloudy, it might rain”
5. Semantic Knowledge	Understanding of meanings and relationships	“A car has wheels”

Declarative Knowledge

- Describes what is.
- Stored in the form of facts, objects, and relationships.

Procedural Knowledge

- Describes how to do something.
- Often encoded in rules, scripts, or algorithms.

Meta-Knowledge

- “Knowledge about knowledge”.
- Useful for evaluating the relevance or accuracy of information.

Heuristic Knowledge

- **Based on experience, intuition, or trial-and-error.**
- **Not always logically correct, but often useful.**

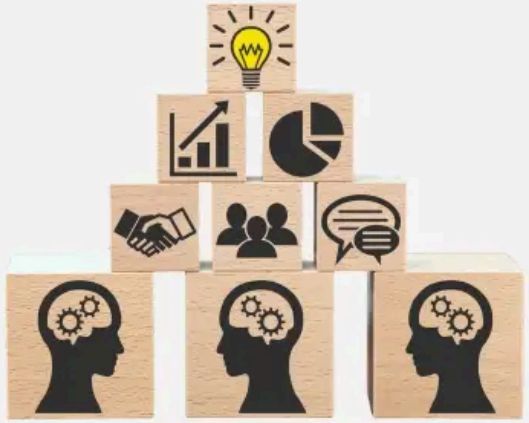
Semantic Knowledge

- **Focuses on meaning and context.**
- **Helps machines understand language, relationships, and categories.**

Methods of Knowledge Representation

AI systems rely on various methods to represent, process, and reason with knowledge. These methods differ in complexity, expressiveness, and suitability for specific tasks.

- **Logic-Based Systems**
- **Structured Representations**
- **Probabilistic Models**
- **Distributed Representations**



Logic-Based Systems

- **Use formal logic to represent rules and facts. Best for deterministic environments requiring high precision.**

Propositional Logic

Represents facts as propositions using logical operators: **AND, OR, NOT**

Example:
IF object is red AND round THEN object might be an apple

✅ **Simple & deterministic**
❌ **Limited expressiveness (no variables or quantifiers)**

First-Order Logic (FOL)

Adds
variables, quantifiers (∀, ∃), and predicates

Example:
∀x (Human(x) → Mortal(x))
“All humans are mortal”

✅ **More expressive**
❌ **Computationally expensive**

✅ **Use Case: Legal AI for contract analysis**

Structured Representations

- **Organize knowledge as networks or hierarchical structures, mimicking human categorization.**

Semantic Networks

Nodes = Concepts;
Edges = Relationships

Example:
Dog → is-a → Animal

✅ **Easy inheritance**
❌ **Lacks formal semantics**

Frames

Group knowledge into
templates or "frames"

Example:
Vehicle Frame →
{wheels: 4, engine: petrol}

✅ **Great for default reasoning**
❌ **Poor at handling exceptions**

Ontologies

Define formal **structure of domain concepts**
Built using standards like **OWL**

Example:
Medical ontology for symptoms, diseases
✅ **Domain standardization & semantic search**
❌ **Complex to design and maintain**

✅ **Use Case: E-commerce, healthcare diagnostics, semantic search**

Probabilistic Models

- **Represent uncertain knowledge using probability.**

Bayesian Networks

Directed graphs with
nodes (variables) and **edges (causal links)**

Example:
Rain → Wet Grass → Slippery Road

✅ **Captures conditional dependencies**
❌ **Requires prior knowledge and probabilities**

Markov Decision Processes (MDPs)

Model decision-making in
dynamic, uncertain environments

Components: **States, Actions, Rewards, Transitions**

✅ **Sequential reasoning**
❌ **Computationally heavy in large spaces**

✅ **Use Case: Weather forecasting, robotics, planning**

Distributed Representations

- **Represent knowledge as dense numerical vectors, capturing patterns and semantics.**

Embeddings

Word2Vec, GloVe

Maps similar words close in vector space

Example:

king - man + woman ≈ queen

✅ **Enables semantic similarity**

❌ **Opaque reasoning (black box)**

Knowledge Graphs

Combine graphs with embeddings

Nodes = Entities, Edges = Relationships

Example:

“Sachin → playsFor → India”

✅ **Enables complex, context-aware reasoning**

❌ **Requires huge datasets and resources**

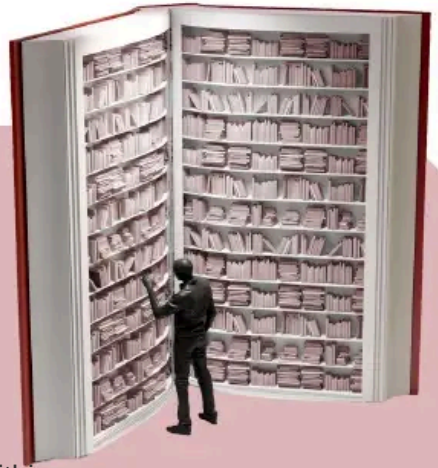
✅ **Use Case: Google Knowledge Graph, chatbots, search engines**

Method	Best For	Limitation	Example Use
Logic-Based	Deterministic reasoning	Rigid, not good with uncertainty	Legal AI
Structured	Hierarchical knowledge	Poor handling of uncertainty	Semantic Web
Probabilistic	Uncertain or noisy domains	Requires probabilistic data	Robotics
Distributed	Large-scale pattern learning	Opaque (black-box) reasoning	Search, NLP

✅ Characteristics of a Good Knowledge Representation System

🔢 Characteristic	📝 Description
1. Representational Adequacy	The system must be capable of representing **all kinds of knowledge** required in the domain. ➤ Can it express facts, rules, procedures, and relationships?
2. Inferential Adequacy	The system should support mechanisms to **derive new knowledge** from existing knowledge. ➤ Can it support reasoning (e.g., deduction, induction)?
3. Inferential Efficiency	Reasoning should be **computationally efficient**—i.e., it should not take too long or too much memory. ➤ Can it find conclusions quickly?
4. Acquisitional Efficiency	The system should make it **easy to acquire new knowledge** from humans, sensors, or other sources. ➤ Can it be updated or expanded with minimal effort?

Applications of KR in AI



- **Expert Systems:** Like MYCIN for medical diagnosis, mimicking human expert decision-making.
- **Natural Language Processing (NLP):** Understanding and responding to human language by interpreting meaning.
- **Robotics:** Enabling robots to interact with and make decisions within their environment.
- **Search Engines:** Google Knowledge Graph uses KR to enhance search results with contextual information.

Summary

Foundational for AI

Enables AI to understand, learn, and reason effectively.

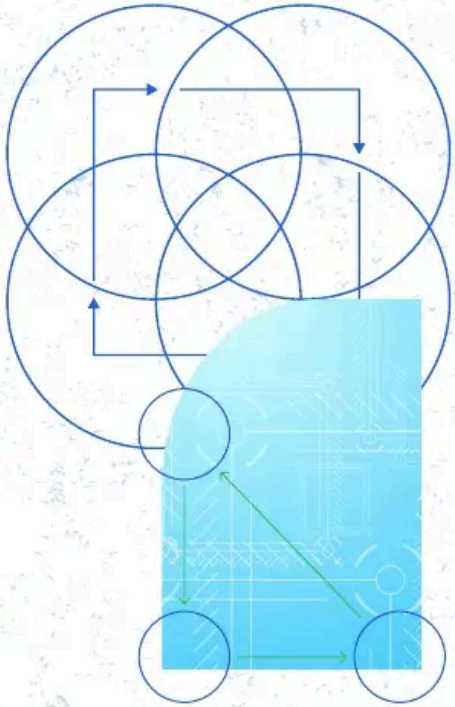
Diverse Applications

Supports a wide range of fields including expert systems, robotics, and NLP.

Tailored Techniques

Different KR techniques are employed based on specific problem domains and complexity.

INTRODUCTION TO MACHINE LEARNING

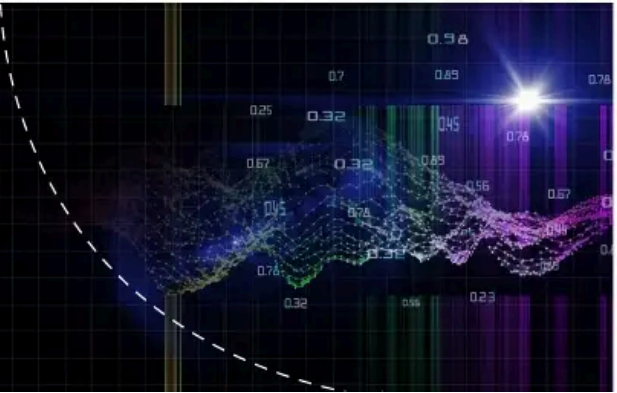


"Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed."

Arthur Samuel (1959)

Understanding Machine Learning

ML = Learn from data + Improve over time



01 Learn from Data

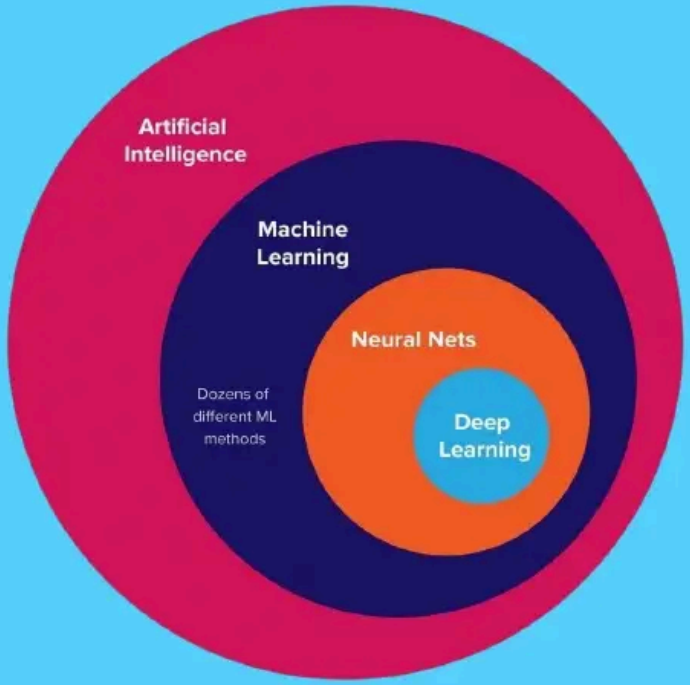
Machine Learning enables systems to **extract insights** from vast datasets, improving their performance through experience.

02 Improve Over Time

As more data is fed into the system, ML algorithms continuously **refine their accuracy** and decision-making capabilities.

03 Automate Decision-Making

Machine Learning facilitates **fast and efficient** decision-making processes, reducing the need for manual intervention and logic rules.


Artificial Intelligence
Machine Learning
Neural Nets
Deep Learning
Dozens of different ML methods

The Need for Machine Learning



01 Challenges of Traditional Programming

Writing rules for every situation is impractical, making it difficult to adapt to new scenarios.

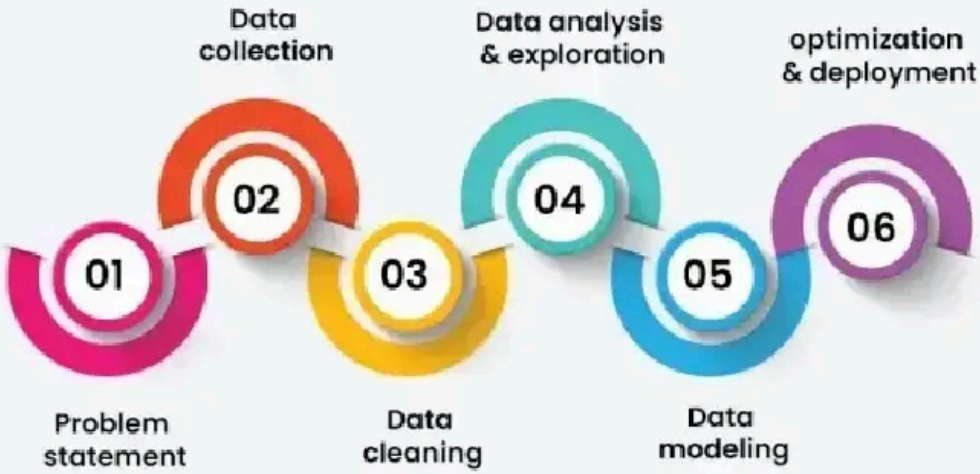
02 Handling Massive Data Volumes

Machine Learning excels at processing and analyzing large datasets that are beyond human capability.

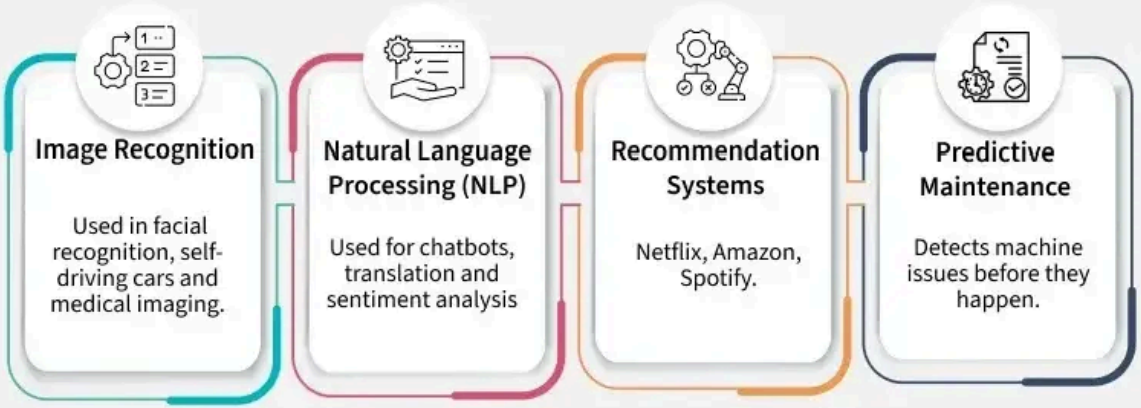
03 Applications in Various Domains

From fraud detection to speech recognition, ML enhances efficiency and accuracy across numerous industries.

How Machine Learning Works



Machine Learning Use Cases



Types of Machine Learning



01 Supervised Learning

Learns from labeled data to predict outcomes. Examples include email spam detection and house price predictions using algorithms like Linear Regression.

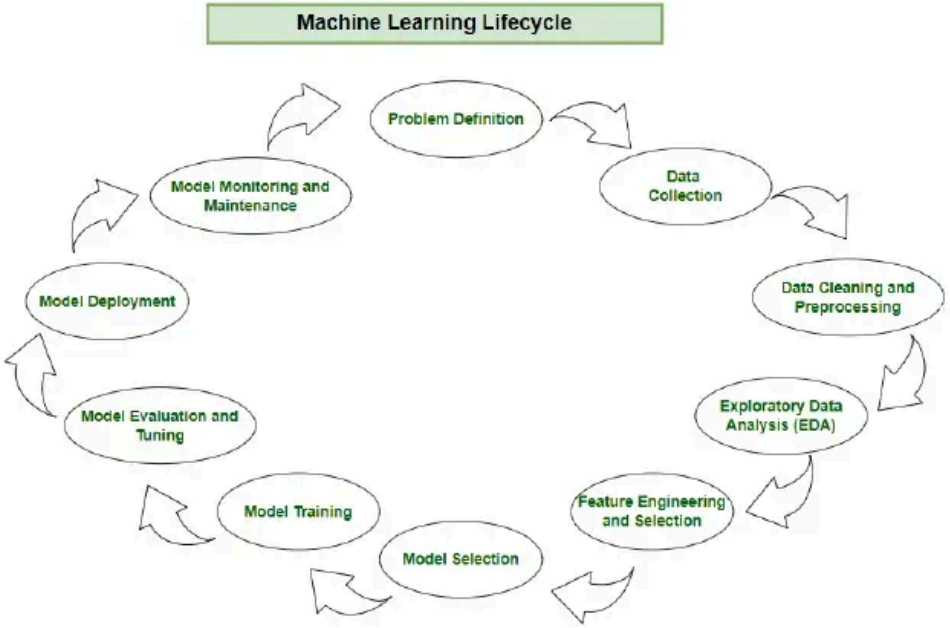
02 Unsupervised Learning

Identifies patterns in data without labels. Applications include customer segmentation and organizing music playlists through algorithms like K-Means Clustering.

03 Reinforcement Learning

Learns strategies through feedback from the environment. Examples include AlphaGo and autonomous driving using algorithms like Q-Learning.

Machine Learning workflow



Key Concepts in ML



🔹 Training Data vs Testing Data

Training Data: The portion of data used to teach the machine learning model.
📌 Used to learn patterns and build the model.

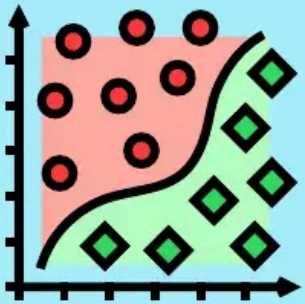
Testing Data: The portion of data used to evaluate the model's performance.
📌 Used to check how well the model generalizes to unseen data.

Key Concepts in ML

🔹 Model

- A model is a mathematical structure that learns from data.
- It captures relationships between inputs (features) and outputs (labels).

📌 Think of it like a smart equation that makes predictions.

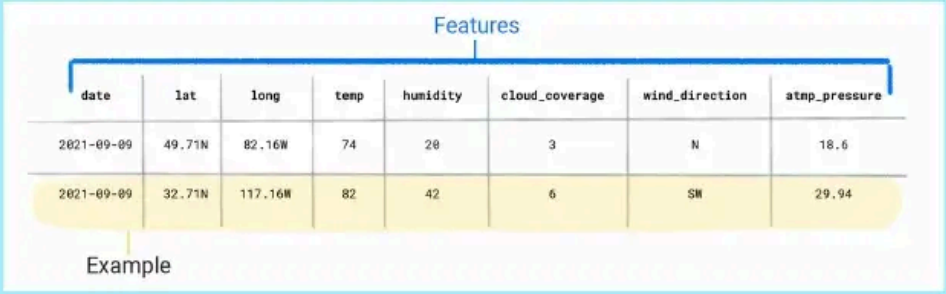


Key Concepts in ML

🔹 Features

- **Features are the input variables used to make predictions.**

📌 **In a table, these are the columns like age, salary, temperature, etc.**

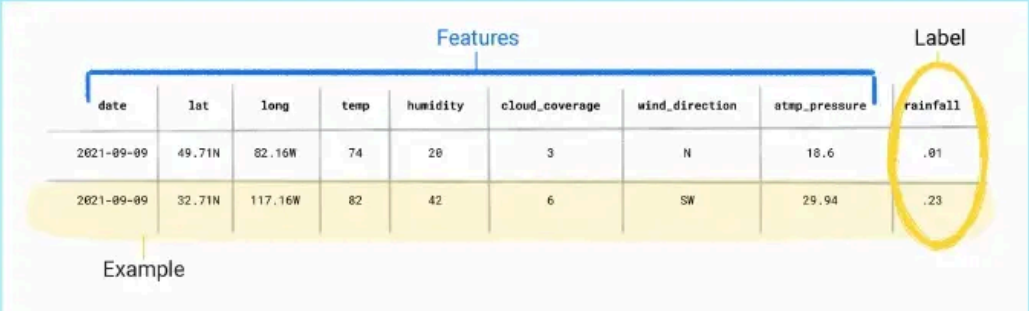


Key Concepts in ML

🔹 Labels

- Labels are the target outputs in supervised learning.

📌 They are the answers the model is trained to predict (e.g., "yes/no", "price", "category").



date	lat	long	temp	humidity	cloud_coverage	wind_direction	atmp_pressure	rainfall
2021-09-09	49.71N	82.16W	74	20	3	N	18.6	.01
2021-09-09	32.71N	117.16W	82	42	6	SW	29.94	.23

Key Concepts in ML

🔹 Overfitting

- The model memorizes the training data perfectly, but performs poorly on new data.
- It fails to generalize.

📉 Too complex → fits even noise in training data.

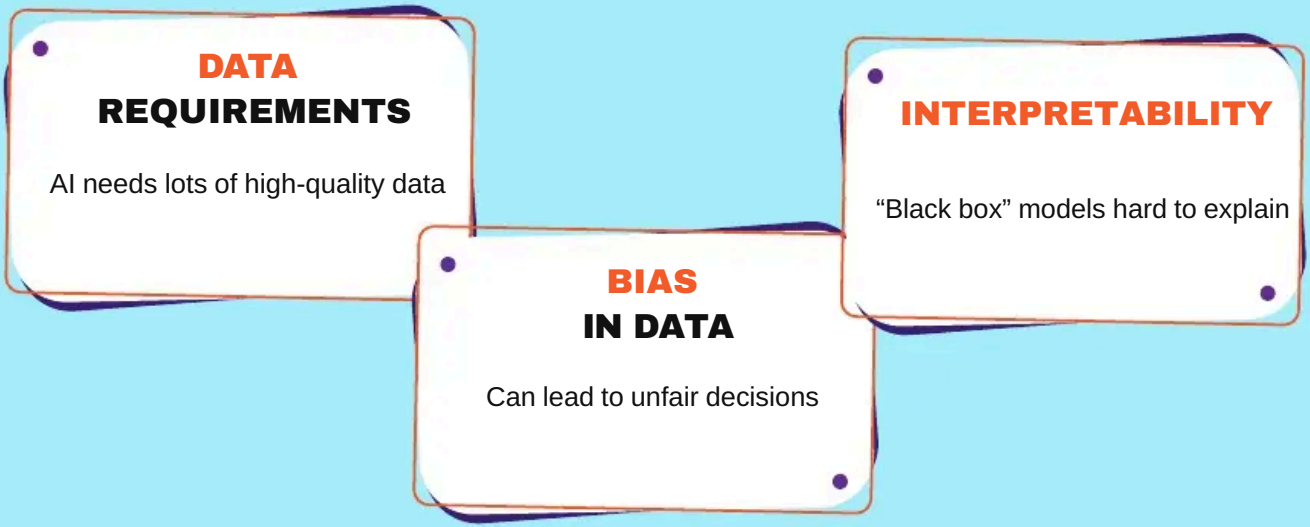
Key Concepts in ML

🔹 Underfitting

- The model is too simple to learn the patterns in the data.
- Performs poorly on both training and testing data.

📉 Too basic → can't capture underlying relationships.

Challenges in AI



Challenges in AI

COMPUTATIONAL COST

Training large models is expensive

SCALABILITY

Difficult to generalize to new tasks

LIMITATIONS OF AI

Limitation	Description
Lacks common sense	Can't reason like humans
No emotional intelligence	Can't understand feelings
Domain-specific	Limited to what it's trained on
Prone to errors	Especially with unseen data

ETHICAL CONCERNS

